

ME-255 A:FLUID MECHANICS I (M)

For

**B.E II YEAR IV SEMESTER
(MECHANICAL ENGINEERING)**



By

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(UNIT -4)

FLUID DYNAMICS

LECTURE 19

Fluid dynamics

"Dynamics means study of state of motion of a body (fluid mass) under the influence of external forces and moments."

Equation of motion

The dynamics of fluid flow is analysed by Newton's second law of motion which relates acceleration of fluid flow with the forces.

Net force acting on a fluid element is given as

$$\sum F = ma$$

Both the forces and acceleration are vector quantities, hence they are resolved in respective directions.

$$\left\{ \begin{array}{l} \sum F_x = ma_x \\ \sum F_y = ma_y \\ \sum F_z = ma_z \end{array} \right\}$$

Forces involved in the fluid motion

In a fluid flow following forces influence the fluid motion

(i) GRAVITY FORCE (F_g)

It is due to weight of fluid and equals mg .

(ii) PRESSURE FORCE (F_p)

Force exerted on fluid mass if there is pressure gradient between two points in the direction of flow.

(iii) VISCOUS FORCE (F_v)

Force due to viscosity of flowing fluid and exists in case of all real fluids.

(iv) TURBULENCE FORCE (F_t)

It is due to turbulence in the flow. It is due to continuous momentum transfer between adjacent layers resulting in development of additional stress (Reynold stress) for flowing fluid.

(v) SURFACE TENSION FORCE (F_s)

It is due to cohesive property of fluid. It is important only when depth of flow is very small.

(vi) COMPRESSIBILITY FORCE (F_c)

It is due to elastic properties of fluid. It is important only for compressible fluids.

Equations of motion

Net force acting on a fluid element is given by

$$\sum F = ma = F_g + F_p + F_v + F_t + F_s + F_c$$

REYNOLD'S EQUATION OF MOTION

In most problems surface tension force and compressibility force are negligible.

$$ma = F_g + F_p + F_v + F_t \rightarrow \text{useful for turbulent flows.}$$

NAVIER STOKES EQUATION OF MOTION

For laminar or viscous flow, turbulent forces are negligible.

$$ma = F_g + F_p + F_v \rightarrow \text{useful for viscous flows.}$$

EULER'S EQUATION OF MOTION

If the fluid is assumed ideal, viscous forces are ignored.

$$ma = F_g + F_p \rightarrow \text{useful for ideal fluid.}$$

LECTURE 20

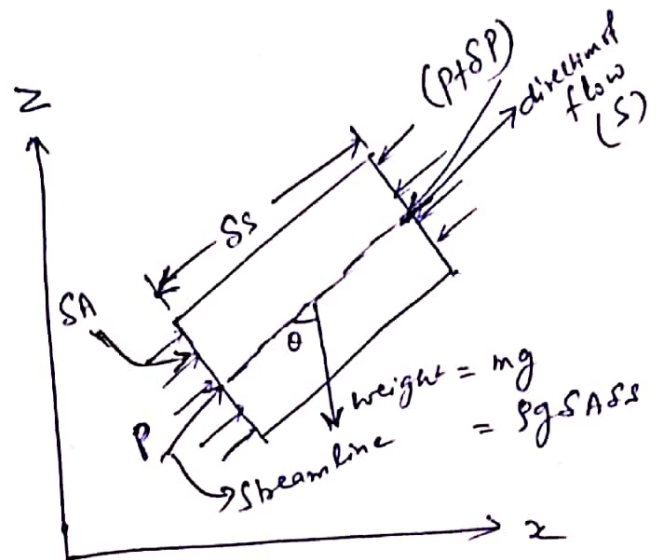
Euler's equation of motion

Consider steady flow of ideal fluid along a stream line.

S = direction of flow along streamline

Consider small element of fluid of cross section area δA and length δs .

θ = angle between direction of flow and line of direction of weight element.



ASSUMPTIONS

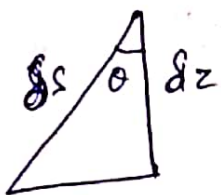
- (i) Fluid is non-viscous.
- (ii) Fluid is incompressible.
- (iii) Flow is steady.
- (iv) Velocity is uniform over cross section.
- (v) Flow is along streamline.
- (vi) Except gravity and pressure forces, no other forces or energy is involved.

Net pressure force acting in the direction of flow

$$P \delta A - (P + \delta P) \delta A = - \delta P \delta A$$

Component of weight of fluid in the direction of flow

$$\begin{aligned} &= - \rho g \delta A \delta s \cos \theta \\ &= - \rho g \delta A \delta s \frac{\delta z}{\delta s} \\ &= - \rho g \delta A \delta z \end{aligned}$$



Velocity of an elementary fluid particle along a streamline is a function of position and time.

$$v = f(s, t)$$

$$dv = \frac{\partial v}{\partial s} ds + \frac{\partial v}{\partial t} dt$$

$$\text{acceleration} = a_s = \frac{dv}{dt} = \frac{\partial v}{\partial s} \frac{ds}{dt} + \frac{\partial v}{\partial t}$$

$$\boxed{a_s = v \frac{dv}{ds}}$$

for steady flow

$$\frac{\partial v}{\partial t} = 0$$

and partial differentials become total differentials.

From Newton's second law of motion

$$\sum F_s = m a_s$$

$$- \rho \delta A - \rho g \delta A \delta z = \rho (\delta A \delta s) v \frac{dv}{ds}$$

dividing both sides by $-\rho \delta A$

$$+ \frac{\delta p}{\rho} + g \delta z + v dv = 0$$

$$\boxed{\frac{dp}{\rho} + g dz + v dv = 0}$$

→ Euler's equation of motion for steady flow

of an ideal fluid along a streamline.

Bernoulli's equation: Integration of Euler's equation for flow along streamline

If the fluid is incompressible i.e. $\rho = \text{constant}$, Then Euler's equation can be integrated.

$$\int \frac{dp}{\rho} + \int g dz + \int v dv = C$$

$$\frac{p}{\rho} + \frac{v^2}{2} + g z = C$$

Pressure head or static head → Velocity head → datum head

$$\frac{P}{\rho g} + \frac{v^2}{2g} + Z = C$$

This is Bernoulli's equation for steady non-viscous, incompressible fluid along a streamline.

"Hence in steady flow of non-viscous, incompressible fluid the sum of pressure head, velocity head and datum head remains constant at every section."

Restatement

"Sum of pressure energy, kinetic energy and potential energy of a steady ideal fluid flow along streamline is constant."

Some facts

— Bernoulli's equation is derived based on assumptions (stated earlier) and corresponds to limitations of its applications.

— For case of a real fluid flow, some energy of flowing fluid is converted to ^{into} heat due to viscosity and turbulent shear. Hence eqn shows an energy loss.

$$\frac{P_1}{\rho g} + \frac{v_1^2}{2g} + z_1 = \frac{P_2}{\rho g} + \frac{v_2^2}{2g} + z_2 + h_L$$

Flow through a confined passage

— Flow through a pipe, converging or diverging duct, a bend, sudden expansion, sudden contraction, the velocity change is governed by equation of continuity. Consequently due to change in velocity, the pressure change is governed by Bernoulli's equation.

LECTURE 21

Applications of Bernoulli's equation

- measurement of discharge through a pipe by venturimeter, flow nozzle and orifice meter.
- measurement of static, stagnation and dynamic pressure and hence velocity by pitot-static tube.

measurement of discharge through a pipe by

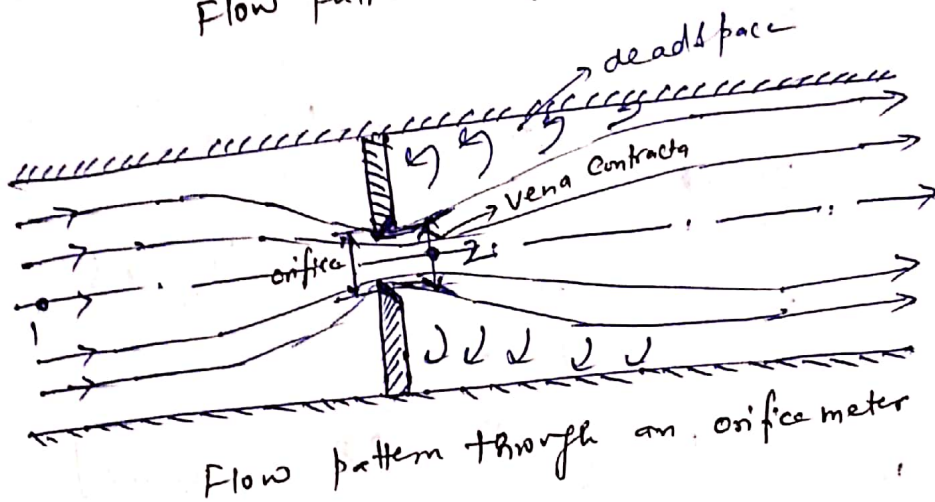
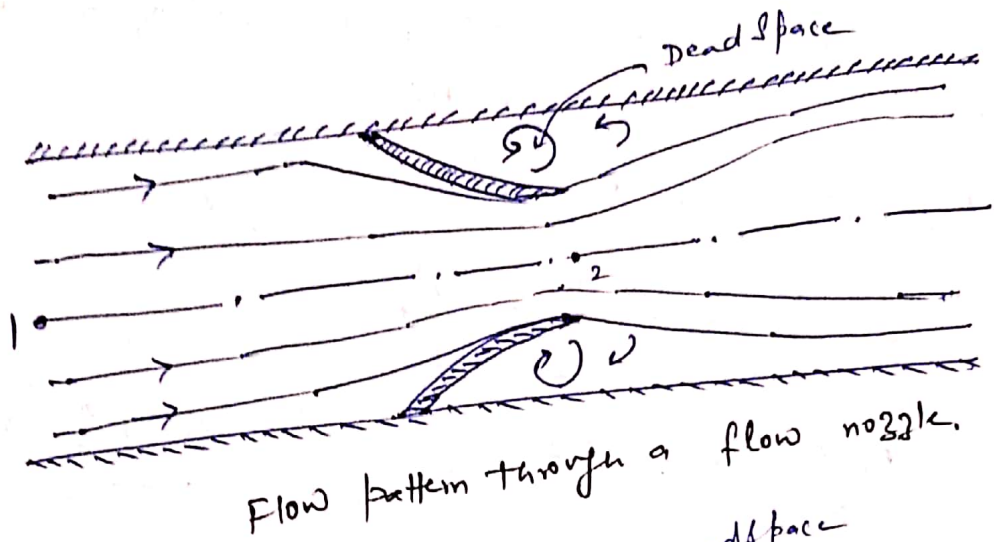
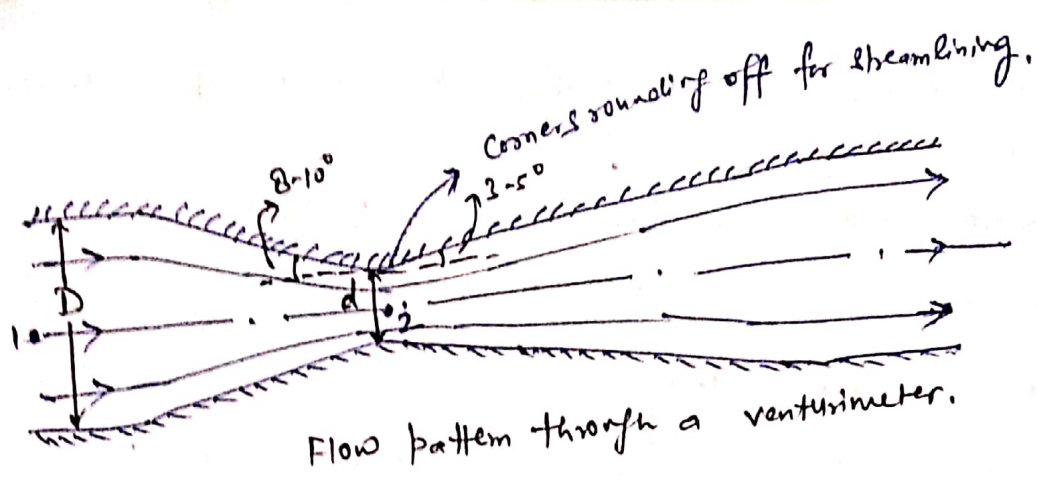
Venturimeter, flow nozzle, orifice meter

"Discharge through a pipe is usually measured by providing a coaxial area contraction within the pipe with a corresponding pressure decrease and causing a pressure difference between two sections. This difference in pressure head is measured by differential manometer and according to Bernoulli's equation is equal to ^{difference in} velocity head between two sections."

Venturimeter

- Venturimeter is a device consisting of short length of gradual convergence, a cylindrical throat and longer length of gradual divergence.
- Semi angle of convergence $20-40^\circ$ and semi-angle of divergence is $3-5^\circ$.
- pressure tapping 1 is provided before convergence section and pressure tapping 2 is provided at the throat.
- pressure difference between two tappings is measured by means of U-tube manometer with water or mercury as suitable manometric fluid.
- Length of throat is equal to d .

d = diameter of throat.



Flow nozzle It is a device through which the contraction of area is brought about by a streamlined convergent nozzle. Here minimum cross-section of flow exists at the exit of the nozzle. Hence one of the pressure tapping is provided at a distance of one diameter upstream the nozzle plate and the other at the nozzle exit.

Orificemeter An orificemeter is a circular plate with an orifice in its centre. The flow adjusts itself such that it contracts until a point downstream of orifice plate (Vena Contracta) and then expands to fill the passage. one pressure tapping is at one diameter upstream of orifice plate and other at vena contracta (approximately at half the pipe diameter downstream of plate).

- In these devices the 2nd tapping is provided at a point where cross section of fluid is minimum.
- Pressure tapping 1 is free stream inlet to device.
- In venturimeter point 2 is at throat.

Expression for rate of flow

Assuming the flow to be inviscid and incompressible and one dimensional between sections 1 and 2, The Continuity equation and Bernoulli's equation may be applied.

Continuity :
eqn $Q = A_1 V_1 = A_2 V_2 \Rightarrow V_1 = \frac{A_2}{A_1} V_2$

Bernoulli's :
eqn $\frac{P_1}{\rho g} + \frac{V_1^2}{2g} + Z_1 = \frac{P_2}{\rho g} + \frac{V_2^2}{2g} + Z_2$

$$\frac{P_1 - P_2}{\rho g} + (Z_1 - Z_2) = \frac{V_2^2 - V_1^2}{2g} = \frac{V_2^2 - \left(\frac{A_2}{A_1}\right)^2 V_2^2}{2g}$$

$$\frac{V_2^2}{2g} \left[1 - \left(\frac{A_2}{A_1}\right)^2 \right] = \frac{P_1 - P_2}{\rho g} + (Z_1 - Z_2)$$

$$V_2^2 = \frac{A_1^2}{A_1^2 - A_2^2} \cdot 2g \left[\frac{P_1 - P_2}{\rho g} + (Z_1 - Z_2) \right]$$

$$V_2 = \frac{A_1}{\sqrt{A_1^2 - A_2^2}} \sqrt{2g \left[\frac{P_1 - P_2}{\rho g} + (Z_1 - Z_2) \right]}$$

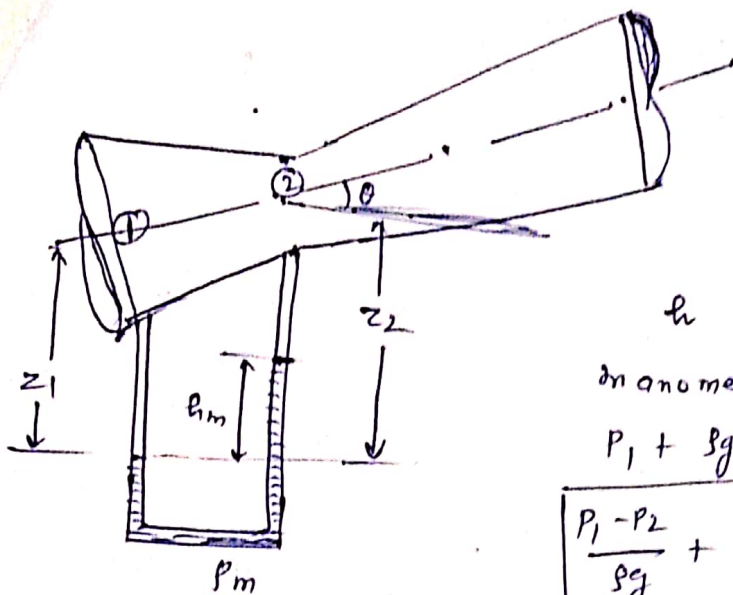
$$Q = A_1 V_1 = A_2 V_2$$

$$Q = \frac{A_1 A_2}{\sqrt{A_1^2 - A_2^2}} \sqrt{2g \left[\frac{P_1 - P_2}{\rho g} + (Z_1 - Z_2) \right]}$$

This is the expression for theoretical discharge.

$$Q = \frac{A_1 A_2}{\sqrt{A_1^2 - A_2^2}} \sqrt{2g h}$$

h = Venturi head = Difference between piezometric heads at sections 1 and 2.



$$\left(\frac{P_1}{\rho g} + z_1 \right) - \left(\frac{P_2}{\rho g} + z_2 \right) = h$$

h can be found out from differential manometer.

$$P_1 + \rho g z_1 - \rho_m g h_m - \rho g (z_2 - h_m) = P_2$$

$$\boxed{\frac{P_1 - P_2}{\rho g} + (z_1 - z_2) = h_m \left(\frac{\rho_m}{\rho} - 1 \right)}$$

- Hence if venturimeter is used to measure discharge in a pipe, which is in inclined position, the value of h may simply be found out by measuring difference in levels of manometric liquid.

Coefficient of discharge

The actual discharge through the flow meter is less than that given by theoretical expression derived on basis of incompressible, inviscid and one dimensional flow.

- In actual practice, flow is not one dimensional.
 - velocity across a cross section does not remain constant. This is due to real flow effects eg. boundary layer formation near the wall and eddy formation in the vicinity of vena contracta (particularly in flow nozzle and orificemeter)
- In orificemeter area of vena contracta is less than the area provided by orifice.
- To take into account wall friction and area contraction effects, a multiplying factor C_d is taken

$$C_d = \text{Coefficient of discharge} = \frac{\text{actual discharge}}{\text{Theoretical discharge}}$$

$$Q_{\text{actual}} = C_d \frac{A_1 A_2}{\sqrt{A_1^2 - A_2^2}} \int 2g h_m \left(\frac{h_m}{s} - 1 \right)$$

Some facts

- In convergent section of venturimeter, the fluid accelerates from inlet section to throat section and then decelerates to fill the passage thereafter. The acceleration is allowed to take place rapidly in small length. However, if retardation was allowed to take place rapidly in small length, the flowing fluid will be separated from boundary walls and eddies are formed which results in excessive energy loss. Hence, divergence cone is made longer. Since separation occurs in divergence cone, this portion is not used for discharge measurement.
- For greater accuracy of pressure difference between inlet and throat section, throat section should be reduced considerably but in the limit of vapour pressure at this section to avoid cavitation phenomenon.
- Generally $d = \frac{1}{2} D$.
- Orificemeter principle is same as of venturimeter but it is a cheaper arrangement. Also where the space is limited, orificemeter may be used for measurement of discharge through pipes. Orificemeter is used for measuring discharge in large diameter pipes.
- Range of C_d

Venturimeter	0.95 - 0.98
Orificemeter	0.6 - 0.65
Flow nozzle	0.7 - 0.8

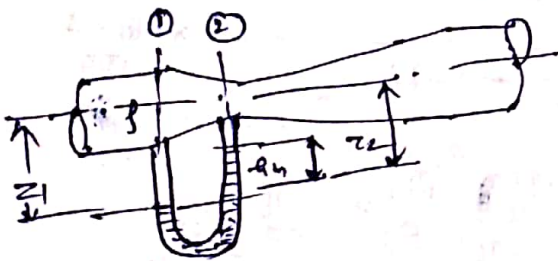
TUTORIAL 6

Tutorial Problem (Solved)

A venturimeter with 200mm diameter at inlet and 100mm throat is laid with axis horizontal and is used for measuring the flow of oil of specific gravity 0.8. The difference of levels in the V-tube differential manometer reads 180 mm of mercury whilst 11.52×10^3 kg of oil is collected in 4 minutes. Calculate coefficient of discharge for venturimeter. Sp. gravity of mercury is 13.6.

Solution: $Q = \frac{A_1 A_2}{\sqrt{A_1^2 - A_2^2}} \sqrt{2gh}$

$h = \text{Venturi head} = \text{difference of piezometric heads at 1 \& 2}$
 $h = \left(\frac{P_1}{\rho g} + z_1 \right) - \left(\frac{P_2}{\rho g} + z_2 \right) = \frac{P_1 - P_2}{\rho g} + (z_1 - z_2)$



$$P_1 + \rho g z_1 = \rho_m g h_m + \rho g (z_2 + h) = P_2$$

$$\frac{P_1 - P_2}{\rho g} + \frac{z_1 - z_2}{g} = h_m \left(\frac{\rho_m}{\rho} - 1 \right)$$

$$A_1 = \frac{\pi}{4} (0.2)^2 = 0.0314 \text{ m}^2$$

$$A_2 = \frac{\pi}{4} (0.1)^2 = 0.00785 \text{ m}^2$$

$$Q =$$

$$\frac{\frac{\pi}{4} \times (0.2)^2 \times \frac{\pi}{4} \times (0.1)^2}{\left[\left(\frac{\pi}{4} (0.2)^2 \right)^2 - \left(\frac{\pi}{4} (0.1)^2 \right)^2 \right]} \sqrt{2 \times 9.8 \times h_m \left(\frac{\rho_m}{\rho} - 1 \right)}$$

$$= \frac{0.0314 \times 0.00785}{\sqrt{(0.0314)^2 - (0.00785)^2}} \times \sqrt{2 \times 9.8 \times 0.18 \times 13.6}$$

$$= 0.06091 \text{ m}^3 \text{ (theoretical)}$$

$$\text{actual} = \frac{11.52 \times 10^3}{0.8 \times 1000 \times 4 \times 60} = 0.06 \text{ m}^3 \text{ (actual)}$$

$$C_d = \frac{\text{Theoretical discharge}}{\text{actual discharge}} = 0.985$$

$$\left(\frac{\rho_m}{\rho} - 1 \right)$$

$$= \left(\frac{13.6}{0.8} - 1 \right)$$

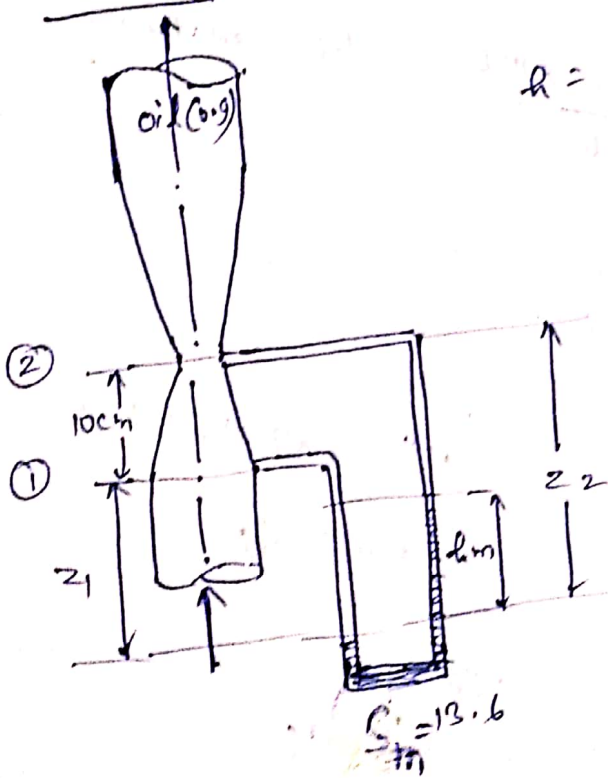
$$= 16.25$$

Tutorial problem (Solved)

An oil of relative density 0.90 flows through a vertical pipe of diameter 10 cm. The flow is measured by a 20 cm x 10 cm venturimeter. The throat is 10 cm above the inlet section. A differential U-tube manometer containing mercury is connected to the throat and the inlet. If $C_d = 0.99$, what's

- the flow for a manometer reading of 9 cm.
- the manometer reading for a flow of 50 L/s?

Solution



$$P_1 + \rho g z_1 - \rho_m g h_m - \rho g (z_2 - h_m) = P_2$$

$$h = \frac{P_1 - P_2}{\rho g} + (z_1 - z_2) = h_m \left(\frac{\rho_m}{\rho} - 1 \right)$$

$$Q = \frac{C_d A_1 A_2}{\sqrt{A_1^2 - A_2^2}} \sqrt{2g h}$$

$$A_1 = \frac{\pi}{4} \times (0.2)^2 = 0.0214 \text{ m}^2$$

$$A_2 = \frac{\pi}{4} \times (0.1)^2 = 0.00785 \text{ m}^2$$

$$\sqrt{A_1^2 - A_2^2} = 0.0204$$

$$h = \frac{0.9 \left(\frac{13.6}{0.9} - 1 \right)}{1.27 \text{ m}}$$

$$(i) \quad Q = \frac{0.99 \times 0.0214 \times 0.00785}{0.0204} \sqrt{1.27 \times 2 \times 9.8}$$

$$= 0.04 \text{ m}^3/\text{s} = 40 \text{ L/s}$$

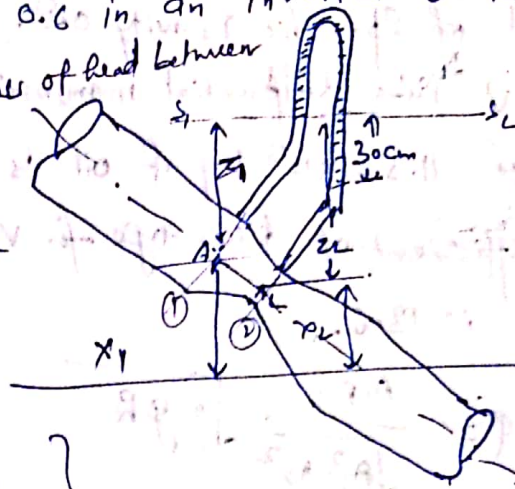
$$(ii) \quad Q = 50 \text{ L/s}$$

$$= 0.05 \text{ m}^3/\text{s}$$

$$\Rightarrow h_m = 0.14 \text{ m} = 14 \text{ cm}$$

Tutorial Problem (Solved)

Find the discharge of water flowing through a pipe 30 cm diameter placed in an inclined position where a Venturimeter is inserted, having a throat diameter of 15 cm. The difference of pressure between main and throat is measured by a liquid of Sp. Gra. 0.6 in an inverted U-tube which gives a reading of 30 cm. The loss of head between main and throat is 0.2 times the kinetic head of the pipe.



Soln.

$$A_1 = \frac{\pi}{4} \times (30)^2$$

$$A_2 = \frac{\pi}{4} \times (15)^2$$

$$P_1 + \rho g z_1 + \rho g (30) + \rho g (z_2 - 30) = P_2$$

$$(P_1 - P_2) + \rho g (z_2 - z_1) = 1000 \times 9.81 \times 30 - 0.6 \times 9.81 \times 30 \times 1000$$

$$\frac{P_1 - P_2}{\rho g} + (z_2 - z_1) = \frac{2(\rho g - \rho_m g)}{\rho g}$$

$$= \frac{2(1000 - 600)}{1000} = 0.8$$

$$= \frac{1177.20}{9.81} = 0.12$$

Bernoulli's

$$\frac{P_1}{\rho g} + \frac{v_1^2}{2g} + z_1 = \frac{P_2}{\rho g} + \frac{v_2^2}{2g} + z_2 + \text{head loss}$$

$$\frac{P_1 - P_2}{\rho g} + (z_1 - z_2) = \frac{v_2^2 - v_1^2}{2g} + 0.2 \times \frac{v_1^2}{2g}$$

$$0.12 = \frac{15.2 v_1^2}{2g}$$

Continuity

$$A_1 v_1 = A_2 v_2$$

$$v_2 = v_1 \times \frac{A_1}{A_2} = v_1 \times \left(\frac{30}{15}\right)^2 = 4v_1$$

$$v_1 = 39.35 \text{ cm/s}$$

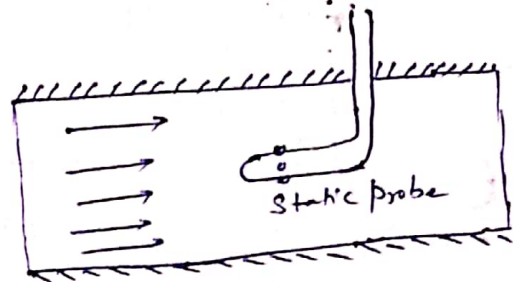
$$Q = A_1 v_1 = \frac{\pi}{4} \times (30)^2 \times (39.35)$$

$$= 0.2782 \text{ m}^3/\text{s}$$

LECTURE 22

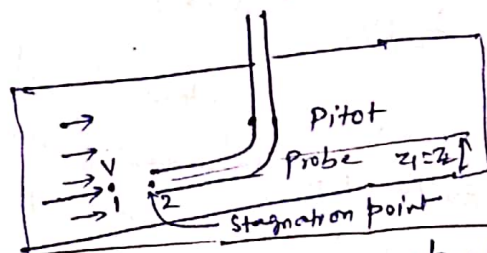
Measurement of static, stagnation and dynamic pressure and velocity

Static pressure in the fluid is measured by a static probe inserted in fluid. The probe has static pressure holes such that the axis of holes is normal to velocity of fluid. There is normally three or more holes on the periphery of the probe and when connected to manometer provide static pressure at the cross section.



STAGNATION PRESSURE at a point in a fluid flow is the total pressure which would result if the fluid were brought to rest isentropically.

A simple device to measure stagnation pressure is a tube with a hole in front inserted in the flow such that velocity is normal to the plane of hole i.e. along the axis of hole. Stagnation point is created by bringing fluid to rest at the nose of the tube and pressure at that point corresponds to stagnation pressure.



$$\text{Stagnation pressure} = \text{Static pressure} + \text{dynamic pressure}$$

Point 1 is taken at which the velocity is V . Point 2 lies immediately at the nose of the tube. This is stagnation point where velocity of flow is reduced to zero.

Applying Bernoulli equation between 1 and 2

$$\frac{P_1}{\rho g} + \frac{V_1^2}{2g} + \frac{z_1}{g} = \frac{P_2}{\rho g} + \frac{V_2^2}{2g} + \frac{z_2}{g}$$

Since $z_1 = z_2$ and $V_2 = 0$ at the stagnation point:

$$\frac{P_1}{\rho g} + \frac{V_1^2}{2g} = \frac{P_2}{\rho g}$$

$$P_2 = \underbrace{P_1}_{\text{Static pressure}} + \underbrace{\frac{1}{2} \rho V_1^2}_{\text{dynamic pressure}}$$

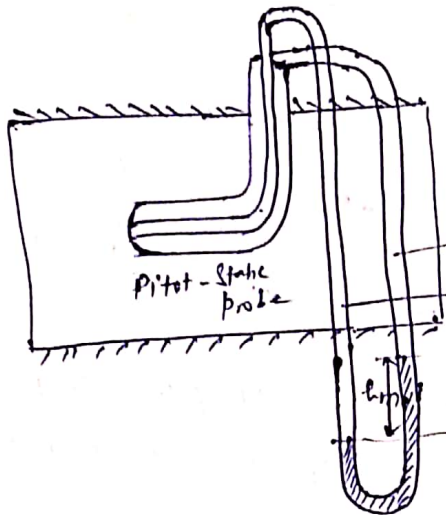
Stagnation pressure

Pitot-static probe measures the difference of the two pressures i.e. stagnation pressure and static pressure using U-tube differential manometer and used to measure velocity of the flow.

$$P_2 = P_1 + \frac{1}{2} \rho v^2$$

$$\frac{v^2}{2g} = \frac{P_2 - P_1}{\rho g} = h_m \left(\frac{\rho_m}{\rho} - 1 \right)$$

$$v = \sqrt{2g h_m \left(\frac{\rho_m}{\rho} - 1 \right)}$$



Static pressure
Stagnation pressure
The whole of the stream impinging the tip may not be brought to rest, so a factor known as coefficient of tube is used to account for this aspect.

$$v = C \sqrt{2g h_m \left(\frac{\rho_m}{\rho} - 1 \right)}$$

$$C = 0.95-1 \text{ (for well designed pitot tube)}$$

Tutorial problem (Solved)

A pipe carries a flow of oil of R.D. = 0.85. A pitot static tube is inserted into the pipe to measure velocity at a point. If the differential mercury oil gauge connected to the pitot-static tube indicates a reading of 4cm, calculate the velocity at that point. Assume the coefficient of pitot tube as 0.99.

Solution

$$v = C \sqrt{2g h_m \left(\frac{\rho_m}{\rho} - 1 \right)}$$

$$= 0.99 \sqrt{2 \times 9.81 \times 0.04 \times \left(\frac{12.6}{0.85} - 1 \right)}$$

$$= \underline{\underline{3.397 \text{ m/s}}}$$

TUTORIAL 7

Tutorial problem (Solved)

A pitot tube is inserted in a pipe of 30 cm diameter. The static pressure of the tube is 10 cm of mercury vacuum. The stagnation pressure at the centre of the pipe recorded by pitot tube is 1.0 N/cm^2 . Calculate rate of flow of water through the pipe, if the mean velocity of the flow is 0.85 times the centre line velocity. Assume coefficient of tube to be 0.98.

Solution

$$\frac{P_{\text{stagnation}}}{\rho g} = \frac{1}{10^{-4} \times 1000 \times 9.81} = 1.019 \text{ m of water.}$$

$$\frac{P_{\text{static}}}{\rho g} = \frac{-10 \times 10^{-2} \times 13.6}{1000 \times 9.81} = -1.36 \text{ m of water}$$

$$\frac{P_{\text{stag}} - P_{\text{static}}}{\rho g} = \frac{1.019 - (-1.36)}{1000 \times 9.81} = 2.379 \text{ m}$$

$$V = \sqrt{2g \left(\frac{P_2 - P_1}{\rho g} \right)} = 0.98 \sqrt{2 \times 9.81 \times 2.379} = 6.695 \text{ m/s}$$

$$\text{mean velocity} = 0.85 \times 6.695 = 5.69 \text{ m/s}$$

$$Q = \frac{\pi}{4} d^2 V_{\text{mean}} = \frac{\pi}{4} \times (0.3)^2 \times 5.69$$

$$Q = 0.402 \text{ m}^3/\text{s}$$

LECTURE 23

Momentum equation

Newton's second law states that the time rate of change of momentum is proportional to applied force and takes place in the direction of force.

Dynamic force applied in x-direction

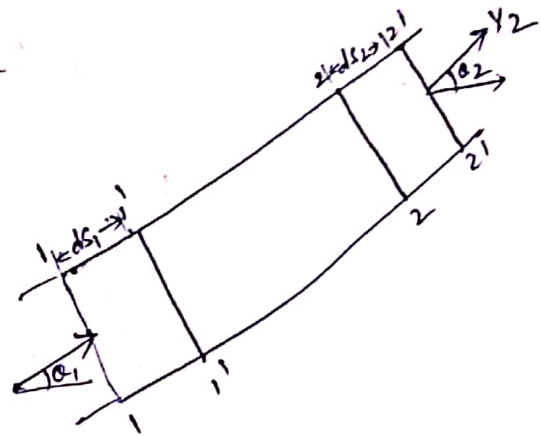
$$F_x = m \frac{dv_x}{dt} \rightarrow \text{Called linear momentum equation.}$$

$$F_x \cdot dt = m \cdot dv_x \rightarrow \text{Called impulse-momentum equation.}$$

Called impulse of applied force

Consider steady flow of fluid through a conduit

Let mass enclosed between sections 1-1 and 2-2 be free body.



In small time dt this free body occupies a new position 1'1' 2'2'

Momentum of fluid entering the section 1-1 in time dt

$$= (\rho_1 A_1 dL_1) v_1 = (\rho_1 A_1 v_1 dt) v_1 \quad v_1 = \frac{dL_1}{dt}$$

Momentum of fluid leaving the section 2-2 in time dt

$$= (\rho_2 A_2 v_2 dt) v_2$$

Change in momentum of the body of fluid

$$= (\rho_2 A_2 v_2 dt) v_2 - (\rho_1 A_1 v_1 dt) v_1$$

Now, net applied force = Rate of change of momentum

$$F = \frac{(\rho_2 A_2 v_2) v_2 \frac{dt}{dt} - (\rho_1 A_1 v_1) v_1 \frac{dt}{dt}}{dt}$$

$$F = \rho_2 (A_2 v_2) v_2 - \rho_1 (A_1 v_1) v_1$$

for steady
incompressible
flow

$$\rho_1 = \rho_2 = \rho$$

$$A_1 v_1 = A_2 v_2 = Q$$

$$F = \rho Q (v_2 - v_1)$$

Since velocity is a vector quantity.

The component of force on fluid body in different directions are

$$F_x = \rho Q (v_{x2} - v_{x1})$$

$$F_y = \rho Q (v_{y2} - v_{y1})$$

$$F_z = \rho Q (v_{z2} - v_{z1})$$

Resultant force on fluid body

$$F = \sqrt{F_x^2 + F_y^2 + F_z^2}$$

Application of momentum equation

Momentum equation can be applied to calculate forces exerted by flowing fluid on the boundaries of flow passage due to change in momentum.

(i) Pipe bends.

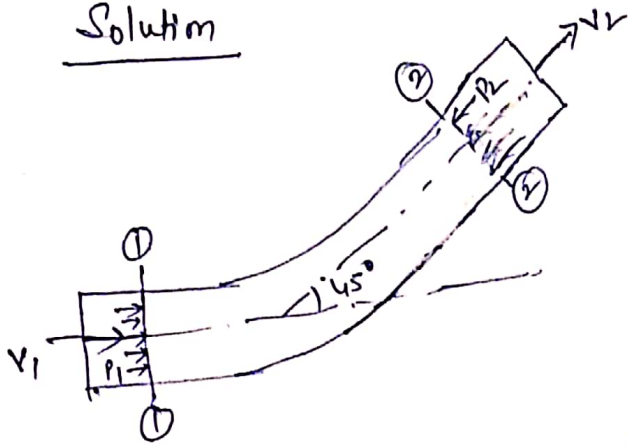
(ii) Fixed and moving Vanes.

TUTORIAL 8

Tutorial problem (Solved)

A 30 cm diameter pipe carries water under a head of 20 meters with a velocity of 3.5 m/s. If the axis of pipe turns through 45° . Find the magnitude and direction of the resultant force on the bend.

Solution



dia. of pipe is same ($A_1 = \frac{\pi}{4} (0.3)^2$)
 $A_1 = A_2 = 0.070685$
 $V_1 = V_2 = 3.5 \text{ m/s}$
 $Q = \frac{\pi}{4} d^2 V_1 = \frac{\pi}{4} \times 0.3^2 \times 3.5$
 $Q = 0.247 \text{ m}^3/\text{s}$

Head = 20m

$\Rightarrow \text{Pressure } P_1 = P_2 = \rho g h$
 $= 1000 \times 9.81 \times 20$
 $= 196200 \text{ N/m}^2$

Force along x axis (F_x)

$F_x = \text{dynamic force} + \text{static force}$

$$= \rho Q (V_1 - V_2 \cos 45^\circ) + P_1 A_1 - P_2 A_2 \cos 45^\circ$$

$$= 1000 \times 0.247 \times (3.5 - 3.5 \times 0.707) + 196200 \times 0.070685 - 196200 \times 0.070685 \times 0.707$$

$$= (864.5 + 196200 \times 0.070685 \times (1 - 0.707))$$

$$= 4217 \text{ N} \quad (\rightarrow)$$

Force along y axis (F_y)

$F_y = \text{dynamic force} + \text{static force}$

$$= \rho Q (0 - V_2 \sin 45^\circ) - P_2 A_2 \sin 45^\circ$$

$$= -1000 \times 0.247 \times 3.5 \times 0.707 - 196200 \times 0.070685 \times 0.707$$

$$= -10416 \text{ N} \quad (\uparrow)$$

$$F = \sqrt{F_x^2 + F_y^2}$$

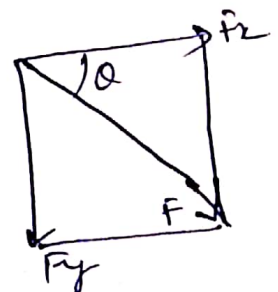
$$= \sqrt{(4217)^2 + (-10416)^2}$$

$$= 11275 \text{ N}$$

$$\tan \theta = F_y / F_x = \frac{-10416}{4217}$$

$$\theta = \tan^{-1}(2.41) = 67.48^\circ$$

an equal and opposite force will be required to hold pipe in position.



LECTURE 24

Flow through orifice and mouth pieces

Orifice

It is a small opening of closed perimeter (eg. circular, rectangular or triangular) on the side or bottom of the tank through which fluid may be discharged.

Mouth piece

is a pipe of short length having length not more than two to three times of its diameter fitted to circular opening.

both are used for measuring discharge of fluid.

Classification of orifice

1. According to Size

Small - If head of liquid from the centre of orifice is more than five times the depth of orifice.
Large - If less

2. According to Shape

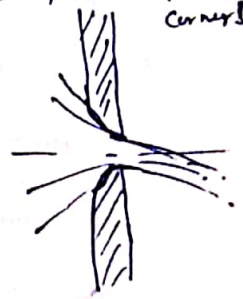
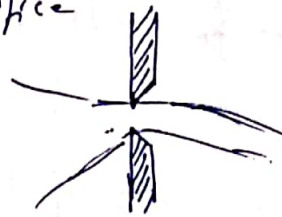
Circular
Rectangular
Square
Triangular

3.

According to Shape of upstream edge

Bell mouthed orifice
i.e. orifice with rounded corners

Sharp edged orifice



In bell mouth orifice liquid contacts the entire inside surface of the orifice.

- In sharp edged orifice the levelled sides are facing down stream and there is minimum contact with the fluid flowing through orifice and minimum frictional effects. It is standard orifice used for measurement of discharge.

4. According to nature of discharge

Orifice discharging free

Drowned or submerged orifices

Fully submerged orifices
Partially submerged orifices

Classification of mouth pieces

1. Shape

Cylindrical, convergent, divergent, Convergent-divergent.

2. Position of mouth piece

External

Internal mouth piece or Bord's mouth piece.

3. Nature of discharge

running full

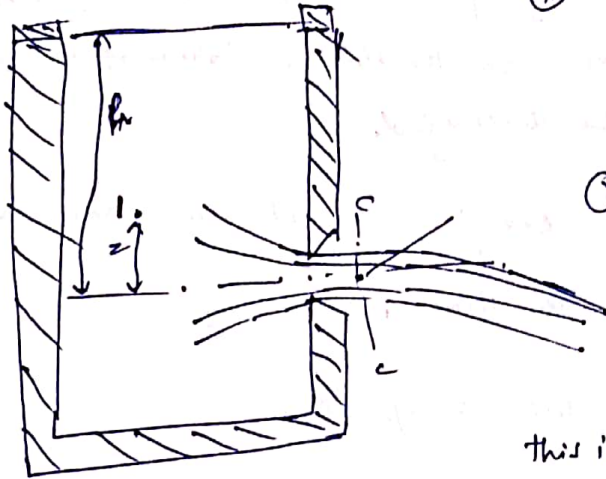
After contraction the jet fills whole the mouth piece

running free

After contraction the mouth piece does not touch the side of mouth piece.

LECTURE 25

Sharp edged orifice discharging free



(*) The jet is discharged to atmosphere and is under the influence of gravity.

(*) Liquid approaching the orifice gradually converges to form a jet whose cross-sectional area is less than that of orifice.

This is because liquid particles lying close to inner wall have a motion along

the wall towards the orifice, which cannot abruptly change the direction at the orifice edge and streamlines continue to converge beyond the orifice upto a certain distance until they become parallel to section cc.

(*) For ideal fluid the parallelism should be approached asymptotically. However due to frictional effects it is only at a short distance from orifice ($\approx 0.5d$) at cc.

(*) Section cc of the jet, at which stream lines are straight and parallel to each other and the jet has minimum cross sectional area is known as Vena-Contracta. Beyond cc the jet diverges again and undergoes deflection due to gravity.

$$Q = q_c \cdot V_2 = q_1 V_1$$

q_1 = cross sectional area of reservoir

for large reservoir $V_1 \approx 0$

$$\frac{p_1}{\rho g} + \frac{v_1^2}{2g} + z_1 = \frac{p_2}{\rho g} + \frac{v_2^2}{2g} + z_2$$

$$\frac{v_2^2}{2g} = \frac{v_1^2}{2g} + h - z_1 + z_2 =$$

$$v_2 = \sqrt{2gh}$$

$$p_1 = p_2 + \rho g(h - z_1)$$

$$p_2 = p_1$$

$$p_1 - p_2 = \rho g(h - z_1)$$

$$\frac{p_1 - p_2}{\rho g} = h - z_1$$

Torricelli's formula.
This gives theoretical velocity

Hydraulic coefficients

1. Coefficient of velocity (C_v) $C_v = \frac{\text{Actual velocity of jet at vena contracta}}{\text{Theoretical velocity}}$

$$C_v = 0.95 - 0.99$$

$$= \frac{V}{\sqrt{2gh}}$$

2. Coefficient of contraction (C_c) $C_c = \frac{\text{Area of jet at vena contracta}}{\text{Area of orifice}}$

$$C_c = 0.61 - 0.69$$

$$= \frac{a_c}{a}$$

3. Coefficient of discharge (C_d) $C_d = \frac{\text{actual discharge}}{\text{Theoretical discharge}}$

$$C_d = 0.61 - 0.65$$

$$= \frac{Q}{Q_{th}} = \frac{Q_{act}}{Q_{th}} = \frac{a_c \cdot V_c}{a \cdot \sqrt{2gh}} = C_v \times C_c$$

$\swarrow$ $\nwarrow$
 theoretical area theoretical velocity

4. Coefficient of resistance (C_r)

$$C_r = \frac{\text{loss of K.E. while flowing through orifice}}{\text{actual K.E. possessed by flowing fluid.}}$$

Notes

C_v : In real fluid flows there is always some loss of energy due to viscous effects and hence actual velocity through the orifice is always less than the theoretical possible velocity.

C_c : Due to convergence of stream lines, area of jet at vena contracta is much less than the area of orifice.

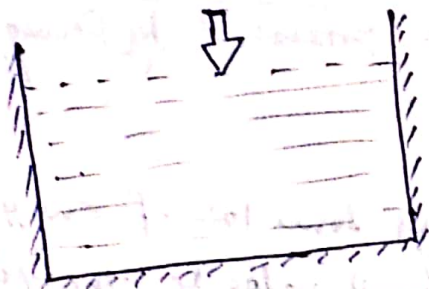
LECTURE 26

Flow over (notches and weirs) → used to measure discharge in open channels

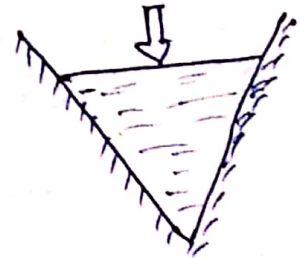
Weir is an obstruction placed in an open channel over which flow occurs. It is in the form of a vertical wall with sharp edge at the top, running all the way across the channel. When the liquid flows over weir, the height of liquid above the tip of sharp edge bears a relationship with the discharge across it.

Notch is a sharp edged device that permits the liquid to go through it. It may be rectangular, triangular, circular or trapezoidal in shape.

[Difference between weir and rectangular notch is that weir runs all the way across the channel while rectangular notch may be as wide as required (over a part of channel).



idealised flow through Rectangular notch



or triangular notch

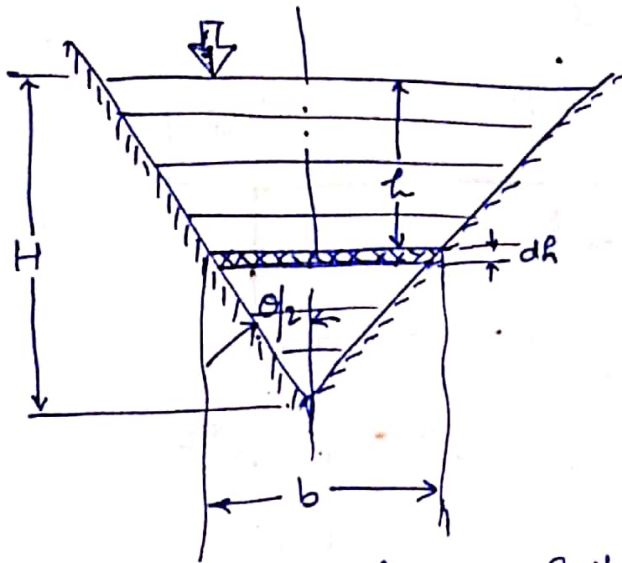


Circular notch



Trapezoidal notch

Discharge through a right angled triangular V-notch



Consider a triangular or V notch with included angle θ and semi-angle $\theta/2$. The liquid flows through it with the level H above the base point.

assume a elementary strip of flow of depth dh located at a depth h below the level of liquid.
discharge through it must be

$$dQ = u \, dA$$

u = velocity of liquid at a point on the strip
 $= \sqrt{2gh}$

$$dA = b \, dh \quad (\text{area of flow through strip})$$

$$dQ = \sqrt{2gh} \, b \, dh$$

$$Q = \int dQ = \int_0^H \sqrt{2gh} \, b \, dh$$

$$= \int_0^H \sqrt{2g} \cdot b h^{1/2} \, dh$$

for triangular notch

$$\tan \theta/2 = \frac{b/2}{H-h}$$

$$b = 2(H-h) \tan \theta/2$$

$$\Rightarrow Q = \int_0^H \sqrt{2g} \cdot 2(H-h) \tan \theta/2 \cdot h^{1/2} \, dh$$

$$= 2 \tan \frac{\theta}{2} \sqrt{2g} \int_0^H (H h^{1/2} - h^{3/2}) \, dh$$

$$\begin{aligned}
 Q &= 2 \tan \frac{\theta}{2} \sqrt{g} \left[\frac{H^{3/2}}{3/2} - \frac{2^{5/2}}{5/2} \right] \\
 &= 2 \tan \frac{\theta}{2} \sqrt{g} \left[\frac{2}{3} H^{5/2} - \frac{2}{5} H^{5/2} \right] \\
 &= 2 \tan \frac{\theta}{2} \sqrt{g} \cdot H^{5/2} \left[\frac{2}{3} - \frac{2}{5} \right]
 \end{aligned}$$

$$\boxed{Q = \frac{8}{15} \tan \frac{\theta}{2} \sqrt{g} H^{5/2}}$$

Actual flow (introduce cd)

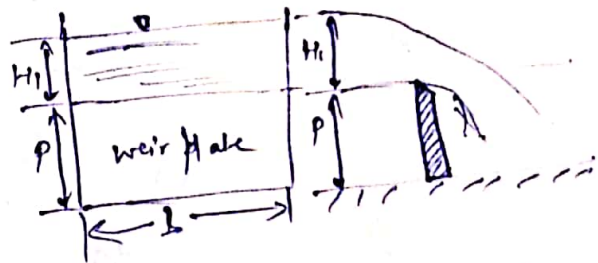
$$Q = \frac{8}{15} C_d \tan \frac{\theta}{2} \sqrt{g} H^{5/2}$$

for right angled V notch.

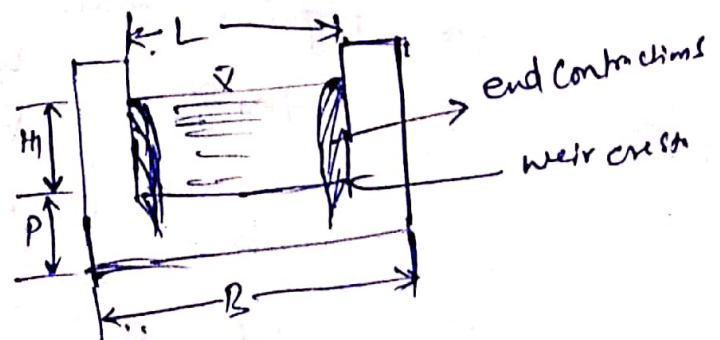
$$\theta = 90^\circ, \quad \frac{\theta}{2} = 45^\circ, \quad \tan \frac{\theta}{2} = 1$$

$$\boxed{Q = \frac{8}{15} C_d \sqrt{g} H^{5/2}}$$

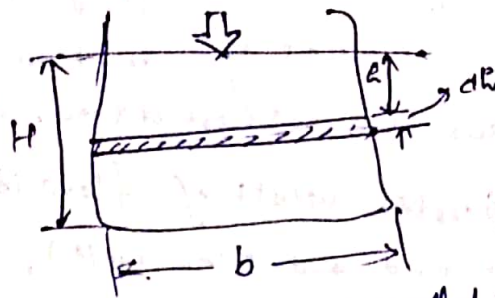
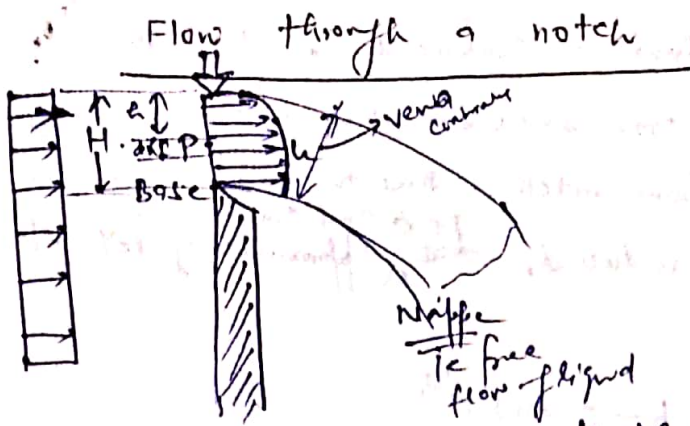
Note! - Suppressed weir
 - when the weir is spanning the full width of a rectangular channel it is called suppressed weir.



- Contracted weir
 - when the length of weir (L) is less than the width of channel (B). It is called contracted weir. Due to end contractions, effective length of weir will be smaller than L.



LECTURE 27



(NOTE: - overflowing stream of liquid in a weir is called Nappe).
 - Top surface over which liquid flows in a weir is called crest).

Flow approaches the notch with a velocity V_1 . Consider a horizontal strip located at a height h below the liquid level.

velocity of liquid at a point on the strip is given by

$$u = \sqrt{2gh}$$

$$\text{flow rate across strip} = dQ = u \cdot dA = u \cdot b \cdot dh$$

$$= \sqrt{2gh} \cdot b \cdot dh$$

$$\text{Total discharge across the notch} = Q = \int dQ$$

Rectangular notch

$$Q = \int_0^H \sqrt{2gh} \cdot b \cdot dh$$

for rectangular notch

$$b = \text{Constant}$$

$$Q = \sqrt{2g} \cdot b \int_0^H h^{1/2} dh = \frac{2}{3} \sqrt{2g} \cdot b H^{3/2}$$

Due to { - minimum area of cross section at vena contracta (the area of flow is less than the area of notch),
 - real flow effects such as side wall boundaries, eddy formation and three dimensional effects,

a factor coefficient of discharge is introduced (C_d)

$$\text{actual discharge } Q = \frac{2}{3} C_d \sqrt{2g} \cdot b H^{3/2}$$

In rectangular notch nappe of flow is contracted at the ends and an area of minimum cross section akin to vena contracta occurs at a short distance from notch. Due to end contraction the effective width of flow is reduced. The end contraction is approximately 10% of head at end (i.e. $0.1H$).

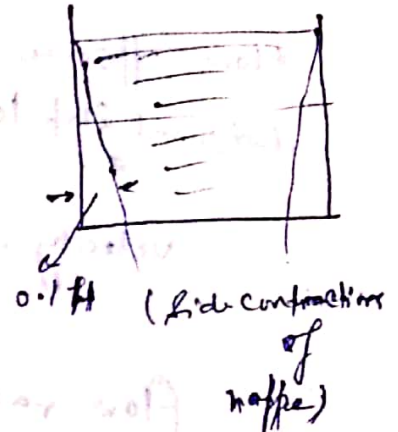
$$Q = \frac{2}{3} C_d \sqrt{g} (b - 0.2H) H^{3/2}$$

reduction in discharge

$$Q = \frac{2}{3} C_d \sqrt{g} (0.2H) H^{3/2}$$

$$= \frac{2}{15} C_d \sqrt{g} H^{5/2}$$

Compare with discharge for a triangular weir



$$\frac{2}{15} C_d \sqrt{g} H^{5/2} = \frac{8}{15} C_d \sqrt{g} H^{5/2} \tan^3 \theta/2$$

$$\tan^3 \theta/2 = 1/4$$

If rectangular weir is enlarged to become trapezoidal weir with side slopes 1 horizontal to 4 vertical. It would give discharge that an uncontracted rectangular weir would have given. This compensated weir is

Called Cipolletti weir.

When a rectangular weir is enlarged to become trapezoidal weir with side slopes 1 horizontal to 4 vertical. It would give discharge that an uncontracted rectangular weir would have given. This compensated weir is called Cipolletti weir.

TUTORIAL 9

Water discharges at a rate of 98.2 litres/s through a 120 mm diameter vertical sharp edged orifice under a constant head of 10 metres. A point, on the jet, measured from the vena contracta of the jet has co-ordinates 4.5 m horizontal and 0.54 m vertical. Find the coefficient of velocity (C_v), contraction (C_c) and discharge (C_d) of the orifice.

Sol.
(Actual discharge)

$$Q = 98.2 \text{ litres/s}$$

$$= 98.2 \times 1000 \text{ cm}^3/\text{s}$$

$$= 98.2 \times 1000 \times 10^{-6} \text{ m}^3/\text{s}$$

$$= 0.0982 \text{ m}^3/\text{s}$$

$$a = \text{area of orifice} = \frac{\pi}{4} \times (12)^2$$

$$= 0.1131 \text{ m}^2$$

$$x = 4.5 \text{ m}$$

$$y = 0.54 \text{ m}$$

$$\begin{aligned} \text{Theoretical velocity} = V &= \sqrt{2gh} \\ &= \sqrt{2 \times 9.81 \times 10} \\ &= 14 \text{ m/s} \end{aligned}$$

$$Q_{th} = 0.1131 \times 14$$

$$= 1.5842 \text{ m}^3/\text{s}$$

$$C_d = \frac{\text{actual discharge}}{\text{Theoretical discharge}} = \frac{0.0982}{1.5842}$$

$$C_d = 0.62$$

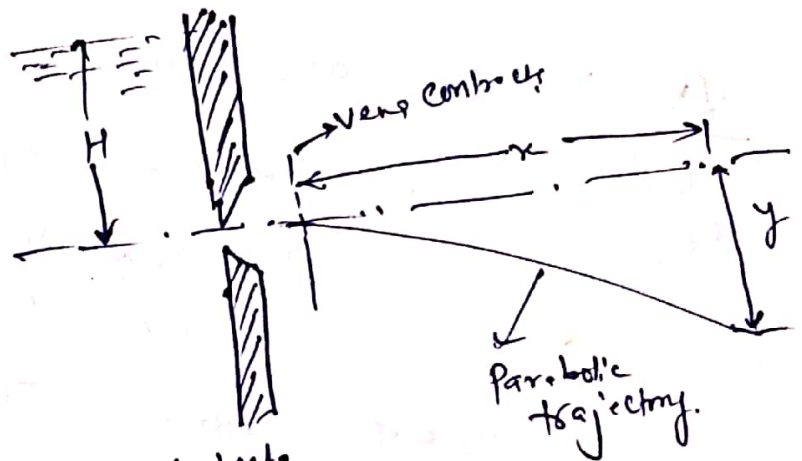
$$C_v = \frac{x}{\sqrt{4yH}} = \frac{6.5}{\sqrt{4 \times 0.54 \times 10}} = 0.968$$

$$C_d = C_v \cdot C_c$$

$$C_c = \frac{0.62}{0.968} = 0.64$$

Determination of C_v

Let x, y are the coordinates of the jet trajectory with origin at vena contracta.



Horizontal distance travelled by jet in time t

$$x = V \cdot t$$

Vertical distance travelled during the same time interval t (due to gravitation)

$$y = \frac{1}{2} g t^2$$

$$y = \frac{1}{2} g \left(\frac{x}{V} \right)^2 \Rightarrow V = \sqrt{\frac{g x^2}{2y}}$$

$$V = C_v \sqrt{2gH} \Rightarrow C_v = \frac{V}{\sqrt{2gH}} = \frac{\sqrt{\frac{g x^2}{2y}}}{\sqrt{2gH}} = \frac{x}{\sqrt{4yH}}$$

$$C_v = \frac{x}{\sqrt{4yH}}$$

Tutorial problem (Solved)

A 4 cm diameter orifice in the vertical side of a tank discharges water. The water surface in the tank is at constant level of 2m. above the centre of orifice. If the head loss in the orifice is 0.20m and the coefficient of contraction can be assumed to be 0.62. estimate: (i) value of coefficient of velocity and coefficient of discharge, (ii) The discharge through the orifice and (iii) The location of the point of impact of the jet on a horizontal plane located 0.5 m below the centre of orifice.

Solution: Theoretical velocity = $V = \sqrt{2gH}$
 $= \sqrt{2 \times 9.8 \times 2} = 6.264 \text{ m/s}$

Head loss across the orifice.

$$0.2 = H - \frac{V_{\text{actual}}^2}{2g} = 2 - \frac{V_{\text{actual}}^2}{2g}$$
$$V_{\text{actual}} = \sqrt{2 \times 9.8 \times (2 - 0.2)}$$
$$= 5.943 \text{ m/s}$$

(i) $C_v = \frac{V_{\text{actual}}}{V_{\text{theoretical}}} = \frac{5.943}{6.264} = \underline{0.949}$

$C_c = 0.62 \Rightarrow C_d = C_v \times C_c = 0.949 \times 0.62 = \underline{0.598}$

(ii) $Q = C_d a \sqrt{2gH} = 0.598 \times \frac{\pi}{4} \times (0.04)^2 \times \sqrt{2 \times 9.8 \times 2}$
 $= 4.73 \times 10^{-3} \text{ m}^3/\text{s} = \underline{4.73 \text{ Litre/s}}$

(iii) $C_v = \frac{x}{\sqrt{4yH}} = \frac{x}{\sqrt{4 \times 0.5 \times 2}} = 0.949$
 $\Rightarrow x = \underline{1.898 \text{ m}}$

Tutorial problem (Solved)

A rectangular weir 0.75 m high and 1.5 m long is to be used for discharging water from a tank under a head of 0.5 m. Estimate the discharge (a) when it's used as a suppressed weir (b) when it's used as a contracted weir. Assume C_d as 0.66

Solution (a) Suppressed weir. $Q = \frac{2}{3} C_d \sqrt{2g} b H^{3/2}$

$$= \frac{2}{3} \times 0.66 \times \sqrt{2 \times 9.81} \times 1.5 \times (0.5)^{3/2}$$

$$= \underline{\underline{1.0336 \text{ m}^3/\text{s}}}$$

(b) For Contracted weir

$$b = b - 0.2H = 1.4 \text{ m}$$

$$Q = \frac{1.0336}{1.5} \times 1.4 = \underline{\underline{0.9647 \text{ m}^3/\text{s}}}$$

Tutorial problem (Solved)

A sharp edged rectangular notch 50 cm broad has been used to measure the discharge estimated as around 20 litres/second. Find percentage error in computing discharge that would be introduced by an error of 2 mm in observing the head over the notch. Take $C_d = 0.62$

Solution $Q = 20 \text{ litres/s} = 20 \times 10^{-3} \text{ m}^3/\text{s}$

$$Q = \frac{2}{3} C_d \sqrt{2g} b H^{3/2}$$

$$= \frac{2}{3} \times 0.62 \times \sqrt{2 \times 9.81} \times 0.5 \times H^{3/2}$$

$$H = 0.0773 \text{ m}$$

differentiating

$$dQ = \frac{2}{3} C_d \sqrt{2g} b \frac{3}{2} H^{1/2} dH$$

$$\frac{dQ}{Q} = \frac{\frac{3}{2} H^{1/2} dH}{H^{3/2}} = \frac{3}{2} \frac{dH}{H}$$

$$dQ = \frac{3}{2} \times \frac{20 \times 10^{-3}}{0.0773} \times 20 \times 10^{-3}$$

$$= 0.776 \times 10^{-3} \text{ m}^3/\text{s}$$

$$= \underline{\underline{0.776 \text{ litres/s}}}$$

$$\therefore \% \text{ error} = \frac{0.776}{20} = \underline{\underline{3.88\%}}$$

Tutorial problem (Solved)

A discharge of 100 l/s is to be measured by a triangular notch of vertex angle 60° . What would be the head over the vertex of the notch? If the accuracy of reading for head is 1 mm , what error in discharge can be expected at this level? Assume $C_d = 0.58$.

Solution

discharge, Q given as (for triangular notch)

$$Q = \frac{8}{15} C_d \sqrt{2g} \tan \theta/2 H^{5/2}$$

$$100 \times 10^{-3} = \frac{8}{15} \times 0.58 \sqrt{2 \times 9.81} \tan 30 H^{5/2}$$

$$Q = K H^{5/2} \quad H = 0.437 \text{ m}$$

$$dQ = K \frac{5}{2} H^{3/2} dH$$

$$\frac{dQ}{Q} = \frac{5}{2} \frac{dH}{H}$$

$$\Rightarrow dQ = \frac{5}{2} \times \frac{1 \times 10^{-3}}{0.437} \times 100 \times 10^{-3}$$

$$= 0.572 \times 10^{-3} \text{ m}^3/\text{s}$$

$$= 0.572 \text{ litres/s}$$

$$\therefore \text{error} = 0.571\%$$

Tutorial problem (Solved)

A Trapezoidal notch has a base width of 0.75 m and a side slope of 1 horizontal to 2 vertical. Calculate discharge over the notch for a head of 0.5 m , Assume $C_d = 0.62$.

Solution: Discharge over rectangular portion

$$Q_1 = \frac{2}{3} C_d \sqrt{2g} b H^{3/2}$$

Discharge over two triangular portion (equivalent to triangular notch)

$$Q_2 = \frac{8}{15} C_d \sqrt{2g} H^{5/2} \tan \theta/2$$

$$Q = Q_1 + Q_2 = C_d \sqrt{2g} H^{3/2} \left[\frac{2}{3} b + \frac{8}{15} H \tan \theta/2 \right]$$

$$= 0.62 \times \sqrt{2 \times 9.81} \times (0.5)^{3/2} \left[\frac{2}{3} \times 0.75 + \frac{8}{15} \times 0.5 \times 1/2 \right]$$

$$= 0.9866 (1.5 + 0.2667)$$

$$Q = 0.6248 \text{ m}^3/\text{s}$$

